

# API Design Principles

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# API Key Design Principles

Best API is one developers can **use without reading the documentation**

## 1. Consistency

- Consistent Naming
- Consistent Patterns ---userDetails, not as user\_D

## 2. Simplicity

- Focus on more use cases
- Intuitive Design -- Minimize complexity, developers understand quickly

# API Key Design Principles

## **3. Security**

- Authentication
- Authorization
- Input Validation
- Rate Limiting

## **4. Performance**

- Caching Strategies
- Pagination --- required post from the set of thousands of posts
- Minimize Payloads
- Minimize round trips

# Protocols influence API design

- HTTP --- Restful APIs, GraphQL
- Websockets --- Realtime APIs
- gRPC --- Microservices
- Protocol choice affects the API structure, performance and capabilities
- API design **must accommodate protocol limitations** and leverage strengths

# API Design Process

- Identify core use cases and user stories
- Define scope and Boundaries
- Determine performance requirements
- Consider Security constraints

# API design approaches

- Topdown approach: start with high level requirements and workflows
  - Best applied during interview
- Bottomup approach: Begin with existing data models and capabilities
  - Best applied if you are already working in company
  - Because you have already data models
- **Contract-First:** Define the API contract before implementation
  - Find how request and response look like
  - Commonly used during interview

# API lifecycle management

- Design
  - Development
  - Deployment and Monitoring
  - Maintenance
  - Deprecation and Retirement
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- **Main phases:** Design, Maintenance, Deprecation and Retirement

- Thank You